plants.plt

Purpose

`plants.plt` holds one parameter set per plant species the model can grow. Each record fills a `plant_db` entry in the `pldb` array. Plant growth, phenology, biomass-to-energy conversion, root and canopy geometry, nutrient uptake, and residue partition fractions all come from this file. HRU plant communities (`plant.ini`) and management operations (`plant.ops`, `landuse.lum`) reference these records by name.

Source

- Reader: src/plant_parm_read.f90
- Type definition: src/plant_data_module.f90, `type plant_db`

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- Lines 3+: one record per plant. List-directed read into `pldb(ic)` (`type plant_db`). The last sub-object is `res_part_fracs` (`type residue_partition_fracs`), which contributes three trailing fields.

The reader has a conditional branch:

```
if (bsn_cc%nam1 == 0) then
  read (104,*,iostat=eof) pldb(ic)
else
  read (104,*,iostat=eof) pldb(ic), pl_class(ic)
end if
```

`bsn_cc%nam1` is a switch in `codes.bsn`. With the default `nam1 = 0`, the standard column list below applies. With `nam1 /= 0`, one extra `pl_class` token (a 25-character name like `row_crop`, `tree`, `grass`) is read at the end of each record.

#	Field	Type	Units	Default	Description
1	plantnm	char(40)	none	“”	plant name (used as the key from other files)
2	typ	char(18)	none	“”	plant category: `warm_annual`, `cold_annual`, `warm_annual_tuber`, `cold_annual_tuber`, `perennial`
3	trig	char(18)	none	“”	phenology trigger: `moisture_gro` or `temp_gro`
4	nfix_co	real	none	0.0	nitrogen fixation coefficient (0.5 for legumes, 0 for non-legumes)
5	days_mat	int	days	110	days to maturity. If 0, heat units run the entire growing season
6	bio_e	real	(kg/ha)/(MJ/m^2)	15.0	biomass-energy ratio
7	hvsti	real	(kg/ha)/(kg/ha)	0.76	harvest index: crop yield over above-ground biomass
8	blai	real	none	5.0	maximum potential leaf area index
9	frgrw1	real	fraction	0.05	fraction of growing season at the 1st point on the optimal LAI curve
10	laimx1	real	fraction	0.05	fraction of maximum LAI at the 1st point on the optimal LAI curve
11	frgrw2	real	fraction	0.4	fraction of growing season at the 2nd point on the optimal LAI curve
12	laimx2	real	fraction	0.95	fraction of maximum LAI at the 2nd point on the optimal LAI curve
13	dlai	real	fraction	0.99	fraction of growing season when LAI starts to decline
14	dlai_rate	real	none	1.0	exponent governing LAI decline rate
15	chtmx	real	m	6.0	maximum canopy height
16	rdmx	real	m	3.5	maximum root depth
17	t_opt	real	deg C	30.0	optimal temperature for plant growth
18	t_base	real	deg C	10.0	minimum (base) temperature for plant growth
19	cnyld	real	kg N / kg yield	0.0015	fraction of nitrogen in yield
20	cpyld	real	kg P / kg yield	0.0003	fraction of phosphorus in yield
21	pltnfr1	real	kg N / kg biomass	0.006	nitrogen uptake parameter 1 (emergence)
22	pltnfr2	real	kg N / kg biomass	0.002	nitrogen uptake parameter 2 (mid-season)
23	pltnfr3	real	kg N / kg biomass	0.0015	nitrogen uptake parameter 3 (maturity)
24	pltpfr1	real	kg P / kg biomass	0.0007	phosphorus uptake parameter 1 (emergence)
25	pltpfr2	real	kg P / kg biomass	0.0004	phosphorus uptake parameter 2 (mid-season)
26	pltpfr3	real	kg P / kg biomass	0.0003	phosphorus uptake parameter 3 (maturity)
27	wsyf	real	(kg/ha)/(kg/ha)	0.01	lower bound of harvest index under water stress
28	usle_c	real	none	0.001	minimum value of the USLE C factor for water erosion
29	gsi	real	m/s	0.002	maximum stomatal conductance
30	vpdfr	real	kPa	4.0	vapor pressure deficit at which `gmaxfr` is valid
31	gmaxfr	real	none	0.75	fraction of maximum stomatal conductance achieved at `vpdfr`
32	wavp	real	none	8.0	rate of decline in radiation use efficiency per unit VPD

#	Field	Type	Units	Default	Description
33	co2hi	real	uL CO2 / L air	660.0	elevated CO2 concentration at the 2nd point on the RUE-CO2 curve
34	bioehi	real	(kg/ha)/(MJ/m^2)	16.0	biomass-energy ratio at the elevated CO2 level (`co2hi`)
35	rsdco_pl	real	none	0.05	plant residue decomposition coefficient
36	alai_min	real	$m^{2/m}2$	0.75	minimum LAI during winter dormant period
37	laixco_tree	real	none	0.3	coefficient used to estimate maximum LAI during tree growth
38	mat_yrs	int	years	10	years to maturity (forced to at least 1 by the reader)
39	bmx_peren	real	metric tons/ha	1000.0	maximum biomass for forest
40	ext_coef	real	none	0.65	light extinction coefficient
41	leaf_tov_min	real	months	12.0	perennial leaf turnover rate with minimum stress
42	leaf_tov_max	real	months	3.0	perennial leaf turnover rate with maximum stress
43	bm_dieoff	real	fraction	0.0	above-ground biomass that dies off at dormancy
44	rsr1	real	fraction	0.0	initial root-to-shoot ratio at the beginning of the growing season
45	rsr2	real	fraction	0.0	root-to-shoot ratio at the end of the growing season
46	pop1	real	plants/m^2	0.0	plant population at the 1st point on the population-LAI curve
47	frlai1	real	fraction	0.0	fraction of maximum LAI at the 1st point on the population-LAI curve
48	pop2	real	plants/m^2	0.0	plant population at the 2nd point on the population-LAI curve
49	frlai2	real	fraction	0.0	fraction of maximum LAI at the 2nd point on the population-LAI curve
50	frsw_gro	real	fraction	0.5	30-day sum of P-PET that triggers monsoon-season growth in tropical plants
51	aeration	real	none	0.2	aeration stress factor
52	rsd_pctcov	real	none	0.0	residue factor for the percent cover equation
53	rsd_covfac	real	none	0.0	residue factor for the surface cover (C factor) equation
54	res_part_fracs%meta_frac	real	fraction	0.85	metabolic fraction of residue
55	res_part_fracs%str_frac	real	fraction	0.15	structural fraction of residue
56	res_part_fracs%lig_frac	real	fraction	0.12	lignin fraction of residue

When `cswat = 2` in `codes.bsn`, the three residue partition fractions are reinterpreted into above- and below-ground metabolic, structural, and lignin pools (`cswat_1_part_fracs`) in `plant_parm_read`. When `cswat /= 2`, the reader overwrites the read values with hard-coded defaults of 0.85, 0.15, and 0.12.

Example

First two records of `refdata/Ames_sub1/plants.plt`:

```
plants.plt
name  plnt_typ  gro_trig  nfix_co  days_mat  bm_e  harv_idx  lai_pot  frac_hu1  lai_max1  frac_hu2  lai_ma
alfa  perennial  temp_gro  0.5  0  20  0.9  4  0.15  0.01  0.5  0.95  0.9  0.5  0.9  3  20  4  0.02  0.003
cana  warm_annual  temp_gro  0  100  34  0.3  4.5  0.15  0.02  0.45  0.95  0.5  0.3  1.3  1.4  21  5  0.03
```

Notes on the example:

- Header column names written by the editor (`plnt_typ`, `gro_trig`, `bm_e`, `harv_idx`, `lai_pot`, `frac_hu1`, `frac_n_yld`, `frac_n_em`, `stcon_max`, `frac_stcon`, `ru_vpd`, `co2_hi`, `bm_e_hi`, `plnt_decomp`, `lai_min`, `bm_tree_acc`, `yrs_mat`, `bm_tree_max`, `ext_co`, `leaf_tov_mn`, `leaf_tov_mx`, `rt_st_beg`, `rt_st_end`, `plnt_pop1`, `frac_lai1`, `frac_sw_gro`, `avg_lig_frac`, `ab_lig_frac`, `bg_lig_frac`) do not match the field names in the source. The reader uses position only.
- The trailing `description` column is not part of `plant_db`. List-directed reads stop after the type is filled, so the description is silently discarded.
- For `alfa`, `days_mat = 0` is the explicit "use heat units for the full season" signal.

Related files

- `codes.bsn`. `bsn_cc%nam1` selects whether `pl_class` is read after each record. `bsn_cc%cswat` selects how residue partition fractions are interpreted.
- `plant.ini`. Lists which plants are present in each HRU at the start of simulation, keyed by `plantnm`.
- `landuse.lum` and `plant.ops`. Reference plant names from this file in management schedules.